

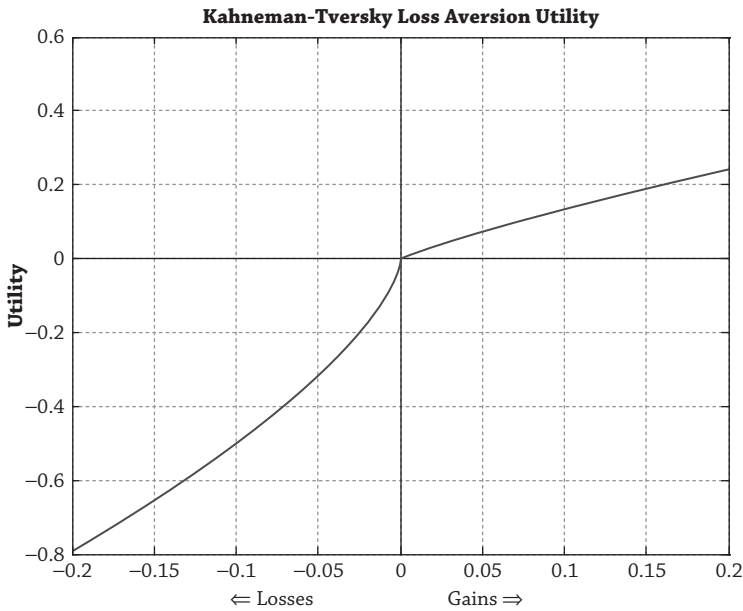


Figure 2.10

People overwhelmingly choose A over B. They are willing to take on some risk to avoid a sure loss.

How investors respond to losses relative to gains is governed by the coefficient of *loss aversion*, which we denote by the parameter λ . Experimental evidence points to estimates of λ around two, so losses are penalized about as twice as much as what gains add in investors’ utilities. Tversky and Kahneman’s (1992) estimate of λ is 2.25.

In Figure 2.11, I plot portfolio weights for optimal investment in the volatility strategy. The graph is produced starting for a regular CRRA investor holding 100% in the volatility strategy. The CRRA investor is a special case of the loss averse investor with no risk asymmetry and $\lambda = 1$. This investor has a risk aversion of $\gamma = 1.68$.²¹ Now we start overweighting losses relative to gains, and we use a reference point of the risk-free rate of $r_f = 1\%$. As the asset owner places greater weight on downside outcomes, he lowers the weight in the volatility strategy. At $\lambda = 2$, he holds approximately only 20% in the volatility strategy and then when he weights downside outcomes more than 2.2 times the weight given gains,

²¹ For comparison, a mean-variance investor with a risk aversion of $\gamma = 3.86$ would hold 100% in the volatility strategy. In this case, mean-variance is not such a good approximation to CRRA—as expected because of the very long left-hand tail of the volatility strategy. Note that that the CRRA investor needs to be more risk tolerant (lower γ) because CRRA utility takes into account the higher moments (see equation (2.1)), whereas mean-variance utility does not.